🔑 **Answer Key — Teacher Copy · Lesson 2-12**

MSTAR PRACTICE WORKSHEET

Lesson 2-12: Multiply Decimals

Grade 6 Mathematics · Standard 6.NOS.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is 3.4×2.6?

(A) 0.884

(B) 8.48

(C) 8.84

(D) 88.4

$34 \times 26 = 884$. There are 2 total decimal places (1 + 1), so place the decimal to get 8.84.

Question 2 SELECTED RESPONSE — CORRECT: B

What is 0.7×1.5?

(A) 0.105

(B) 1.05

(C) 1.50

(D) 10.5

$7 \times 15 = 105$. There are 2 total decimal places (1 + 1), so place the decimal to get 1.05.

Question 3 SELECTED RESPONSE — CORRECT: C

What is 5.3×4?

(A) 2.12

(B) 20.12

(C) 21.2

(D) 212

> $53 \times 4 = 212$. There is 1 total decimal place, so place the decimal to get 21.2.

Question 4 SELECTED RESPONSE — CORRECT: B

Which estimate is closest to 4.8×3.2?

(A) $4 \times 3 = 12$

(B) $5 \times 3 = 15$

(C) $5 \times 4 = 20$

(D) $48 \times 32 = 1{,}536$

> Round 4.8 to 5 and 3.2 to 3. The estimate $5 \times 3 = 15$ is closest to the exact answer 15.36.

Question 5 SELECTED RESPONSE — CORRECT: C

A car travels 55.5 miles per hour for 2.4 hours. How far does it go?

(A) 13.32 miles

(B) 57.9 miles

(C) 133.2 miles

(D) 1,332 miles

> $555 \times 24 = 13{,}320$. Total decimal places: $1 + 1 = 2$. Answer: $133.20 = 133.2$ miles.

Question 6 SELECTED RESPONSE — CORRECT: C

A customer orders 3 taco baskets at $4.25 each. What is the total before tax?

- A $1.275
- B $12.15
- C $12.75
- D $127.50

$3 \times 425 = 1{,}275$. The factor $4.25 has 2 decimal places, so $3 \times \$4.25 = \$12.75$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Find 2.4×0.5.

- (A) 0.12
- (B) 1.2
- (C) 12
- (D) 120

> Multiply as whole numbers: $24 \times 5 = 120$. The factors have $1 + 1 = 2$ decimal places in total, so the product is 1.20, which is 1.2.

- **A:** The point moved three places instead of two. Count one place in 2.4 and one in 0.5.
- **C:** 24 x 5 = 120 is the whole-number product. The two factors hold 2 decimal places between them, so the point moves two places left.
- **D:** That is the whole-number product 24 x 5 with no decimal point placed at all.

Part B — correct answer: C

How is the number of decimal places in the product determined?

- (A) Use the number of decimal places in the larger factor.
- (B) Multiply the number of decimal places in the two factors.
- (C) Add the number of decimal places in the two factors.
- (D) The product always has one decimal place.

> 2.4 has 1 decimal place and 0.5 has 1, so the product has $1 + 1 = 2$ decimal places before trailing zeros are dropped.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about multiplying decimals:

- [x] A. Multiply the factors as if they were whole numbers first.
- [x] B. The total number of decimal places in the factors gives the number in the product.
- [x] C. Multiplying by a number less than 1 gives a product smaller than the other factor.
- [ ] D. The decimal points must be lined up before multiplying.
- [x] E. Estimating first is a good way to check the placement of the decimal point.

> A, B, C, and E are true. D is false — lining up decimal points is required for ADDING and subtracting, not for multiplying.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A driver buys 9.5 gallons of gas at $3.40 per gallon. Show how to find the total cost. Use estimation to check if your answer is reasonable.

Model answer: Total cost is gallons times price per gallon, so the setup is 9.5 × $3.40. Multiplying 95 × 34 = 3,230 and giving the product 1 + 1 = 2 decimal places gives $32.30. Rounding to 10 gallons at about $3 each predicts roughly $30, so $32.30 is reasonable.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
